

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0824

Elwood 230 kV
September 03, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0824. The Interconnection Customer proposes a 56.2 MW wind facility interconnecting at the Elwood 230 kV substation in the Illinois area. The request was received on July 11, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 235 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,531,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hillcrest-Segs 5 100 KV L	Dekalb-Richland 500 KV	135.9	354	7.2
Landman Crk-24Th & Gardenale Substation 230 KV	Dekalb-Richland Substation 500 KV L	134.4	475	9.1
Farmersville-Kendallville 100 KV	Jones Chapel-Dekalb Substation 500 KV L	130.5	252	5.3
43Rd St Terminal-Joliet 29 230 KV	Boca Raton-Pueblo Xtra 500 KV	115.1	816	25.0
Ege-Cuba City 100 KV	Dekalb-Jones Chapel 500 KV	112.2	251	5.3
24T & Gardendale-Sidney A Murray Jr Hydroelectric 230 KV	Dekalb-Richland 500 KV Line	109.5	605	9.1
University-Concord 230 KV Ckt 1	Concord-University 230 KV Circuit 2	107.3	767	12.1
Adirondack 500-230 kV Transformer Bank	East Methuen 500/230 kV Transformer Bank	104.6	1,026	5.1
Concord-University Substation 230 KV	Concord-University 230 KV #1	103.8	793	12.1
Bluebonnet-Russell Fork 230 KV	Dekalb Substation-Jones Chapel 500 KV	103.5	1,597	32.0
Muskogee-Montpelier Electric Generating Station 500 KV	Dekalb-Richland Substation 500 KV	102.3	2,801	20.4
Ege-Golden Gate 100 KV	Jones Chapel-Dekalb 500 KV	101.5	265	5.3
Lonetre Substation-Sabanera 500 KV	Muskogee-Montpelier Electric Generating Sta 500 KV	101.2	2,866	10.8
St. Lawrence-James De Young 230 KV	East Methuen Substation-Lonetree 500 KV Line	100.7	1,026	6.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Napoleon Peaking Station-Richland 230 KV	Troy Energy Llc-Richland 500 KV	100.5	920	13.1
Hebo-Barrett 230 KV	Hebo-Winthrop 230 KV L	100.3	560	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hillcrest-Segs 5 100 KV	Rebuild line with higher-capacity conductor	\$4,606,000	0% (\$24,000)
Landman Crk-24Th & Gardenale Substation 230 KV	Rebuild line with higher-capacity conductor	\$34,048,000	0% (\$178,000)
Farmersville-Kendallville 100 KV	Rebuild line with higher-capacity conductor	\$4,430,000	0% (\$22,000)
43Rd St Terminal-Joliet 29 230 KV	Rebuild line with higher-capacity conductor	\$3,684,000	2% (\$62,000)
Ege-Cuba City 100 KV L	Rebuild line with higher-capacity conductor	\$5,919,000	0% (\$30,000)
Sidney A Murray Jr Hydroelectric Substation-24Th & Gardendale 230 KV	Reconductor line	\$5,861,000	0% (\$31,000)
Concord-University 230 KV Cir. 1	Reconductor line	\$7,631,000	0% (\$37,000)
Adirondack 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$32,626,000	0% (\$169,000)
University-Concord 230 KV Circuit 2	Reconductor line	\$7,163,000	0% (\$34,000)
Bluebonnet-Russell Fork 230 KV	Reconductor line	\$24,123,000	0% (\$125,000)
Montpelier Electric Generating Sta-Muskogee 500 KV Line	Reconductor line	\$59,753,000	0% (\$309,000)
Ege-Golden Gate 100 KV	Reconductor line	\$1,110,000	0% (\$6,000)
Lonetree-Sabanera 500 KV	Reconductor line	\$56,371,000	0% (\$287,000)
St. Lawrence-James De Young 230 KV	Reconductor line	\$22,102,000	0% (\$114,000)
Napoleon Peaking Station-Richland 230 KV	Reconductor line	\$16,010,000	0% (\$84,000)
Hebo-Barrett 230 KV	Reconductor line	\$3,707,000	0% (\$19,000)
	Total allocated to project		\$1,531,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.